

Bachelor thesis

Multimodal Human Activity Recognition Using a Foot-Mounted Edge Sensor Platform

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Task description:

The goal of this bachelor thesis is to design and evaluate a multimodal Human Activity Recognition (HAR) system using the OnlyFeet platform, a low-cost, foot-mounted data logger integrating camera, microphone, IMU, and time-of-flight (ToF) sensors. Unlike traditional wearable HAR systems that rely primarily on inertial sensing, this work focuses on sensor fusion at the foot, enabling richer context awareness such as terrain, obstacles, and environmental cues. The student will use the existing OnlyFeet prototype to record synchronized multimodal sensor data. Based on this data, preprocessing pipelines and feature representations for each modality will be developed. Different sensor fusion strategies (early, mid, or late fusion) will be implemented and evaluated with respect to classification performance and robustness to missing or noisy modalities. The final system will be trained and evaluated offline, with a particular focus on understanding the contribution of each modality and the trade-offs between model complexity and performance.

Key Tasks:

- Conduct a Literature Review related work on multimodal HAR, foot-mounted sensing, and sensor fusion strategies.
- Record a labeled multi-user multimodal dataset using the OnlyFeet prototype.
- Design preprocessing and feature extraction pipelines for IMU, vision, audio, and ToF data.
- Implement and compare different sensor fusion approaches for HAR.
- Evaluate the system using standard metrics (e.g., macro-F1, confusion matrices).
- Analyze modality importance and robustness to partial sensor failure.